



THE UNIVERSITY OF HONG KONG 香港大學
faculty of architecture 建築學院



n2cHK iLab | @HKURBAN
the smart construction lab

空間信息技術賦能文化遺產保護與傳承學術論壇



澳門科技大學
MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY



人文藝術學院
Faculty of Humanities and Arts

混凝土文物建築探地雷達應用初探 Application of Ground-Penetrating Radar in Concrete Heritage Buildings: A Pilot

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5 Sept 2025



The background of the slide is a photograph of a large, ornate building with a clock tower, likely a university or government building. The building is light-colored with many windows and columns. In the foreground, there are lush green palm trees and other tropical vegetation. The sky is a clear, bright blue.

Section 1

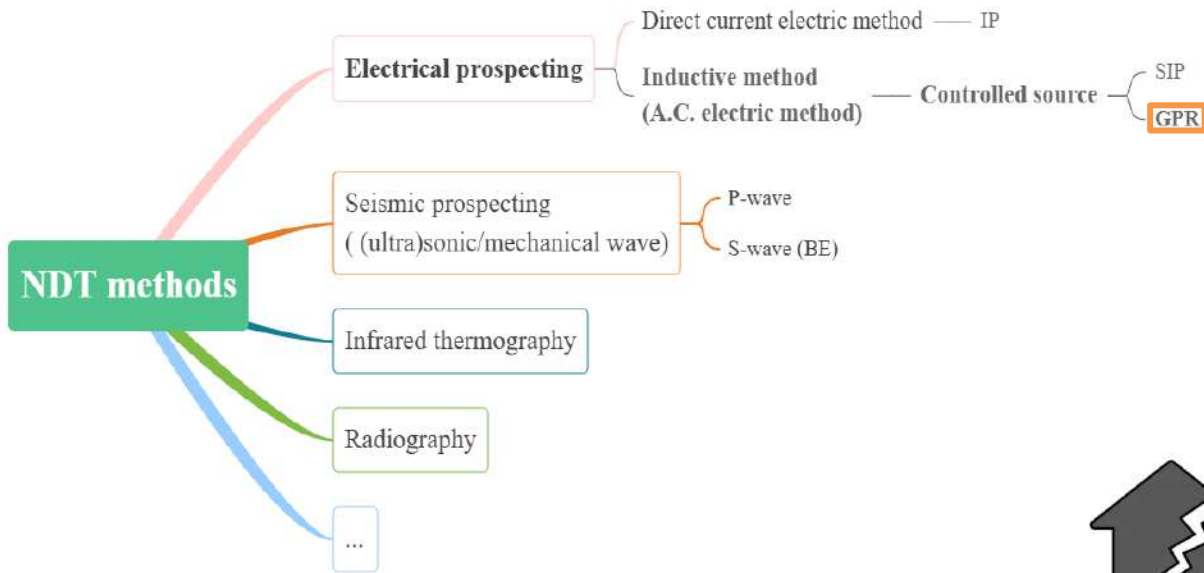
INTRODUCTION TO GPR



Introduction to NDT

iLab ♦ Testing for diagnosis by ‘building doctors’

NDCHK ♦ Nondestructive testing (NDT)



Heritage diagnosis



As-built detection



Concrete cracks



Aging components



Rebar deformation



1 Seeing through concrete

iLab Inside of reinforced concrete 混凝土

- ▣ Rebar 钢筋
- ▣ Connectors 连接件
- ▣ Aggregate 骨材
- ▣ Pipes (with air, water, cable)

◆ Surveying / testing for concrete defects /corrosion 缺陷腐蚀

- ▣ [Destructive] Penetrating / probe 穿刺、探针、破坏
- ▣ [Non-destructive] Ultra-sonic pulse 超声波
- ▣ [Non-destructive] Ground-penetrating-radar (GPR) 探地雷达
 - Polarized high-frequency 10 MHz to 2.6 GHz radio wave
 - E.g., a pipe conductor return a hyperbola in radar image
- ▣ Radioactive (say, X-ray) X光

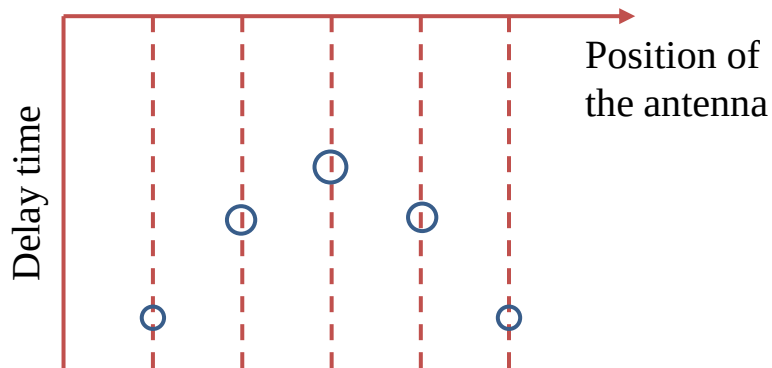
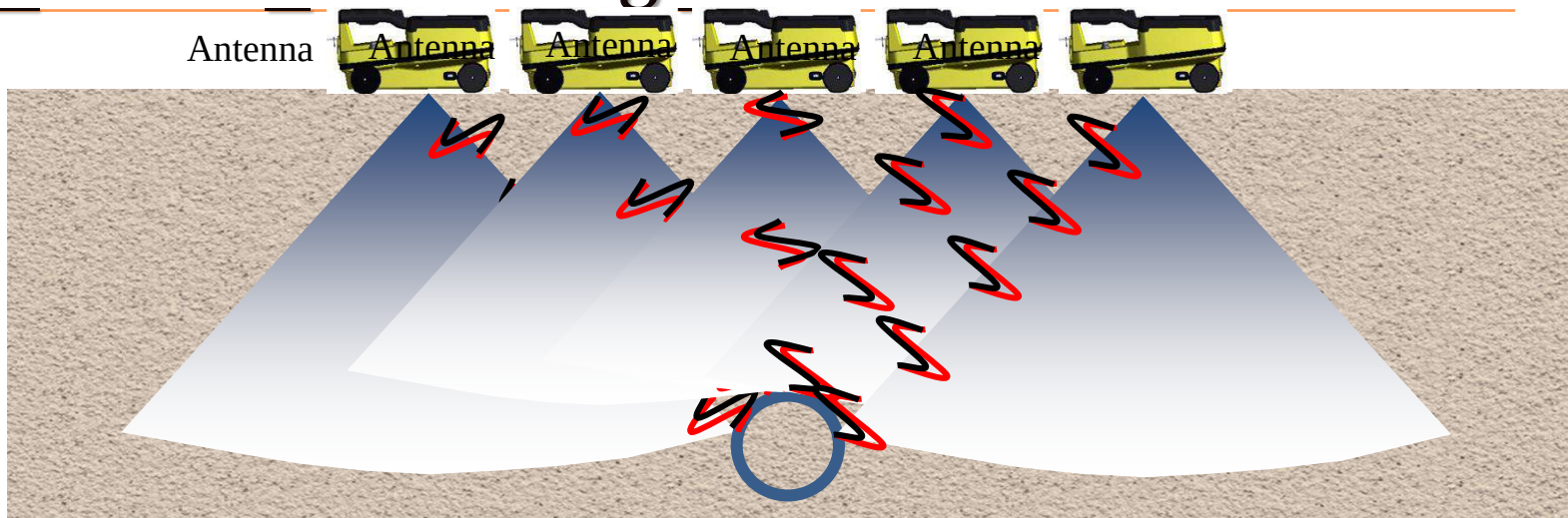




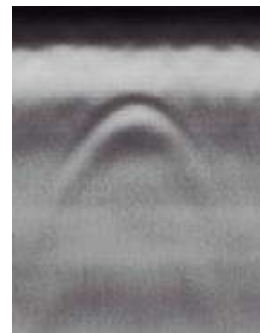
iLab

ADCH

GPR: Ground Penetrating Radar



Processing data

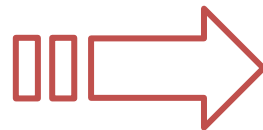
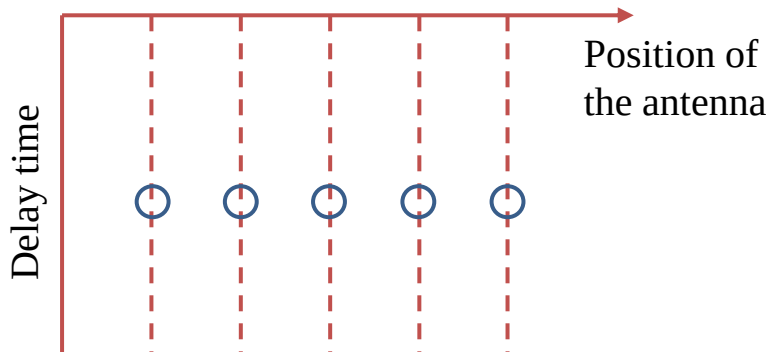
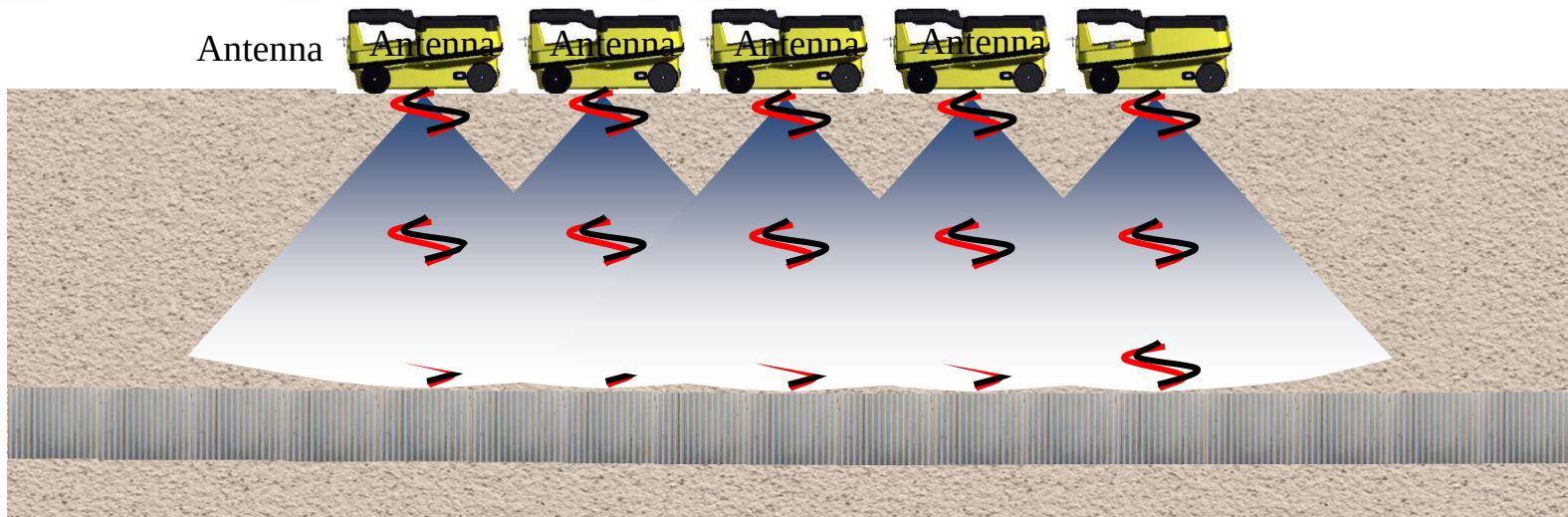




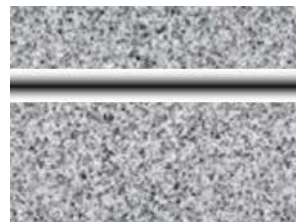
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ADCH

GPR: Ground Penetrating Radar (cont.)



Processing data





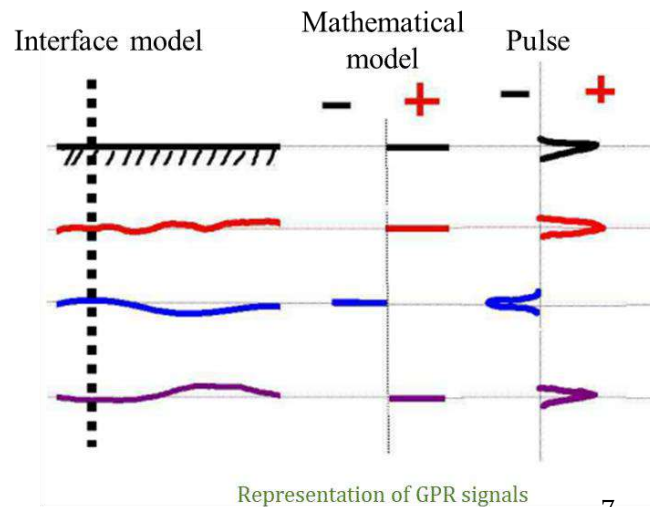
1 GPR signals

iLab

◆ GPR signal property

- ▣ electromagnetic signals
- ▣ signal amplitude decays with depth
- ▣ dependent on **relative permittivity** (or **dielectric constant**) (ϵ_r)
 - insulator to store electric energy in an electrical field

	Material	Common DCR
Reinforced concrete Wall	Concrete	6-8
	Rebar	Near ideal conductor
	Sand (Dry)	2.5-3
Geological Materials	Clay (Wet)	8-15
	Granite	5-8
	Limestone	7-9
	Dolomite	6.8-8
	Sandstone (Wet)	6
	Mudstone (Wet)	7





1. Traditional GPR image

iLab

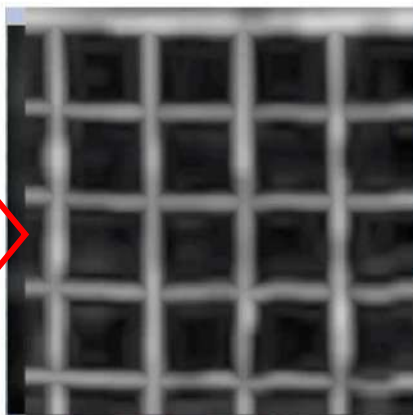
◆ GPR image /C-scans

ADCHK

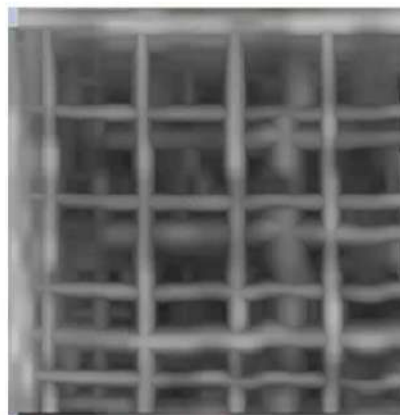
□ Light = conductor 白=导体

□ Dark = insulator

□ Gray = between



Standard GPR Visualisation



C-thru Visualisation



(Sources: Authors; Leica)



1 Limitations and our development

iLab

◆ Limitations in traditional GPR processing

NDCHK

- ▣ **Noise** from reflections/ringing by steel/rebar/power lines.
- ▣ **Ambiguity** in target identification: Similar pattern of root/pipe
- ▣ Depth estimation **errors**:
 - Relies on assumed or measured EM wave velocity
 - errors in velocity lead to incorrect depth calculations.
- ▣ **Clutter** and multiple reflections
 - Complex subsurface leads overlapping signals that are difficult to read/separate.

◆ We design a web AR tool to partly solve these

- ▣ Noise filtering, 3D measurement, clutter filtering



1 Interface functions

iLab

ADCH

✂ GPR-AR

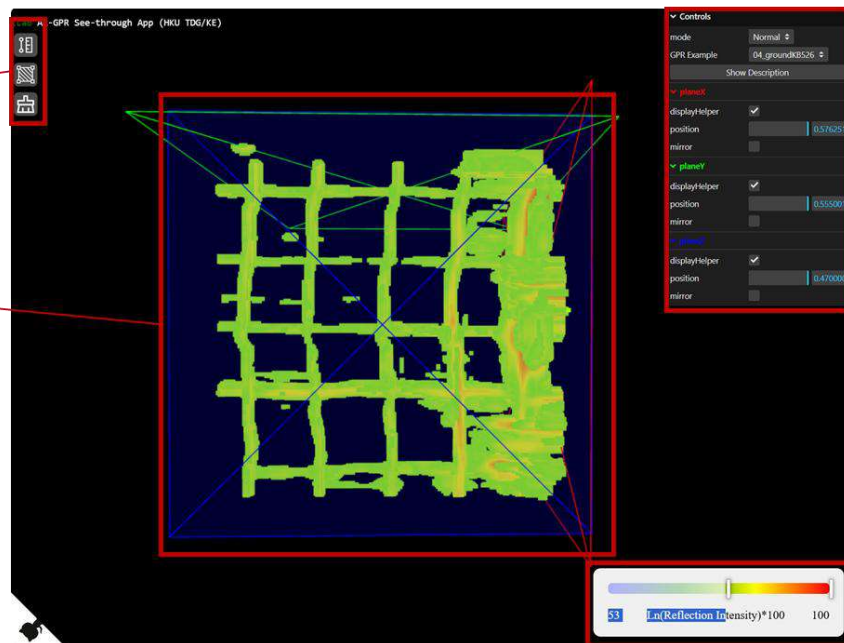
✂ GPR-AR (gprar.netlify.app) is a visualization platform for GPR scanning results

Tools:

- to measure length and area

Scan result:

- the blue plane is the scanning plane normally
- controlling the mouse to move and zoom to realize the 3D visualization



Control panel:

- to choose the mode and example
- to adjust the reference plane

Adjustment Bar:

- the visualization is displayed by the logarithm of reflection intensity
- large difference values indicate different materials



iLab

ADCHK

1 Introduction to device and our GPR-AR



The background of the slide is a photograph of a large, multi-story heritage building with a prominent clock tower on the right side. The building features classical architectural elements like columns and arches. In the foreground, there are lush green trees and plants, some with reddish-brown foliage on the left. The overall scene is bright and sunny.

Section 2

A CASE OF CONCRETE HERITAGE BUILDING



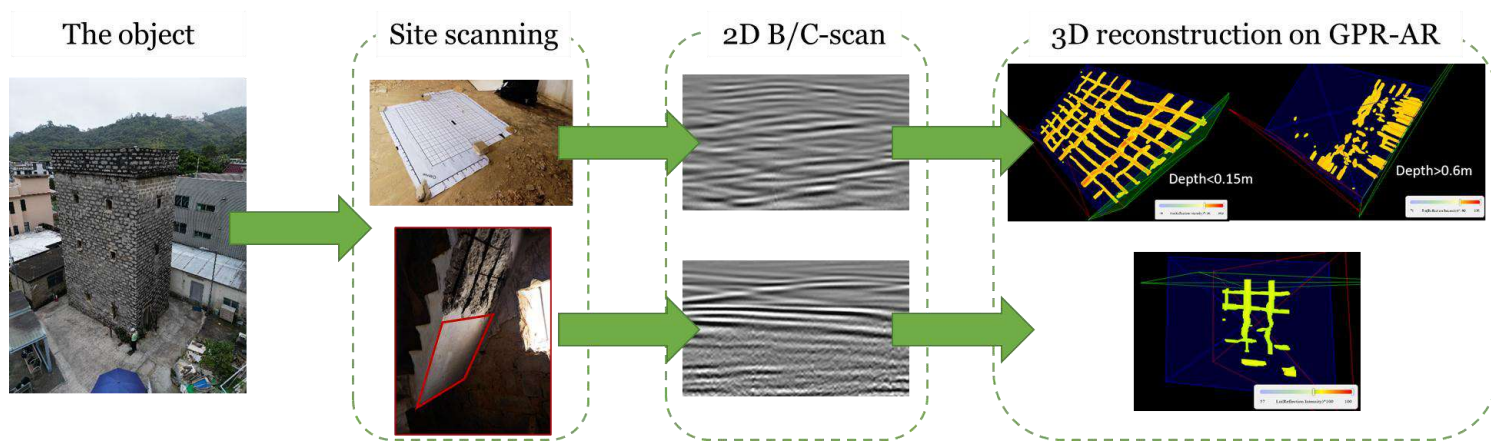
2.1 The Watchtower, Luk Tei Tong

iLab

◆ Watchtower

ADCHK

- ▣ Luk Tei Tong (鹿地塘) Village, Mei Wo, Lantau
- ▣ Around 1942 by the Tsangs (曾) villagers
- ▣ Defense against pirates, bandits and the Japanese.
- ▣ Grade of historic buildings: III
- ▣ Converted for residential use in the 1950s





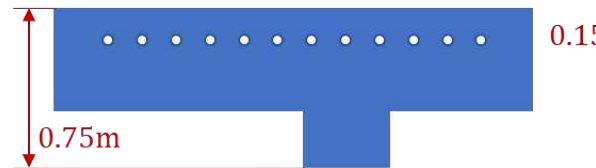
2.3 Result 1 on the ground

iLab

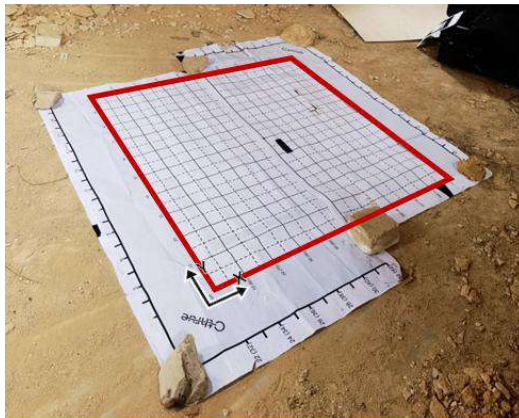
◆ Muiwo-1, Ground

NDCH

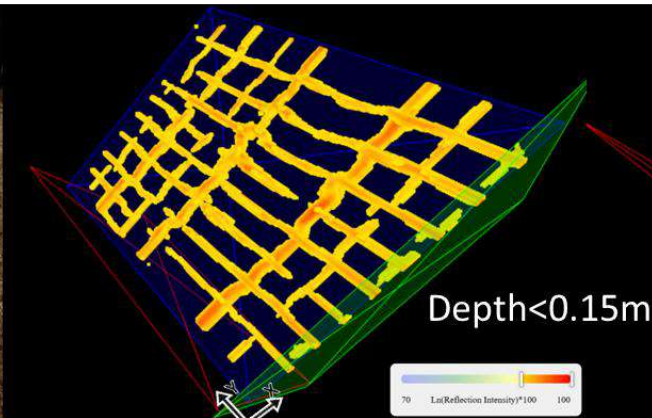
- ▣ A rebar grid can be found at depth $< 0.15\text{m}$; clearly visible.
- ▣ A beam can be detected at a depth range of $[0.6, 0.75]\text{m}$.



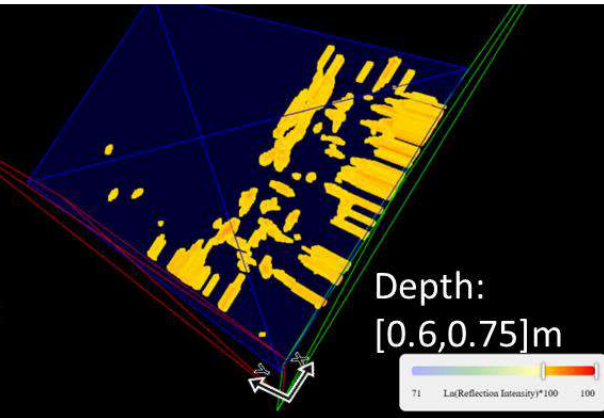
Cross-sectional schematic diagram



Scanning site



3D visualization of the rebar grid



3D visualization of the beam

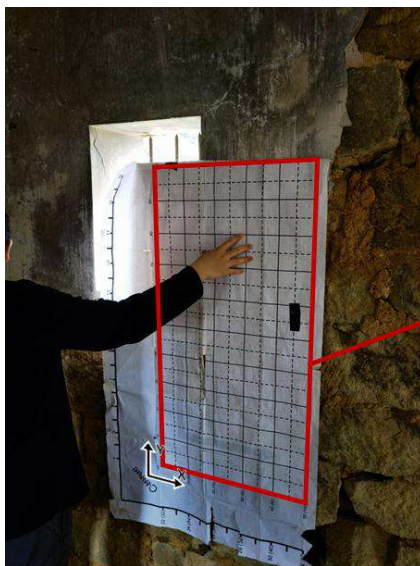


Result 2 on wall

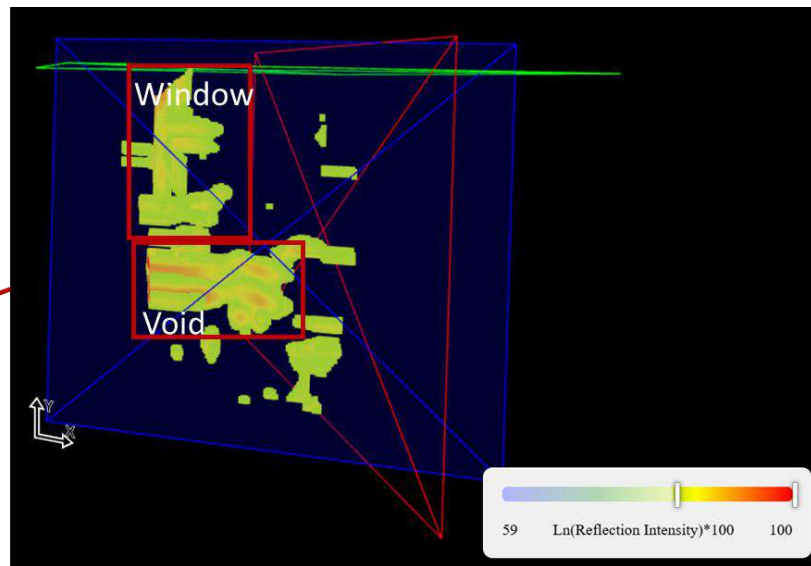
iLab

◆ Muiwo-2, Wall

- NDCH
- ▣ No visible rebar, and the void between the stone is seen.
- ▣ Site: 2/F ([here](#))



Scanning site



3D visualization



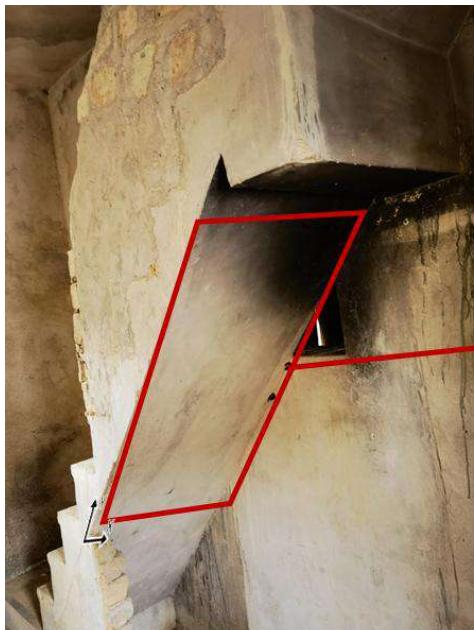
Result 3 on stairs

iLab

◆ Muiwo-3, Stairs

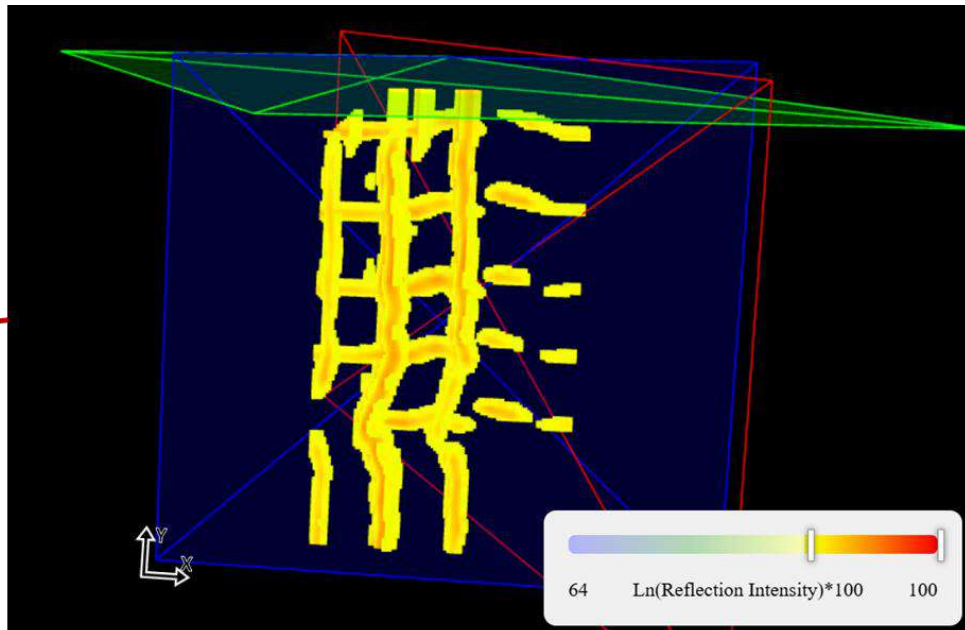
NDCHK

- ▣ A rebar grid can be measured at depth= 0.06m.



Xue: Snaring

Scanning site



3D visualization of the rebar grid



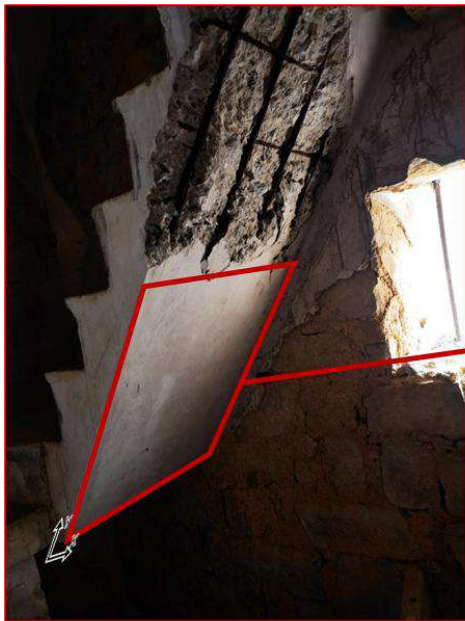
Result 3 on stairs 2

iLab

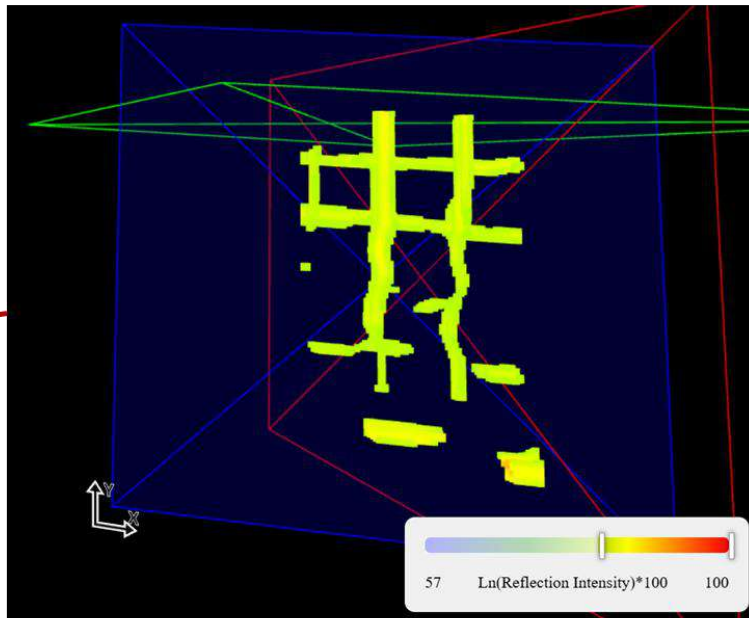
◆ Muiwo-4, Stair #2

NDCH

- ▣ A rebar grid can be observed at depth=0.06m
- ▣ but the scan signal is a little worse than that of Muiwo-3.



Scanning site



3D visualization of the rebar grid

Section 3

WRAP UP



3 Summary

iLab

◆ GPR

NDCH

- ▣ An NDT technology
- ▣ But has many limitations
- ◆ We designed a web AR tool in post-processing GPR data
- ◆ Piloted in a concrete heritage building
 - ▣ Confirmed depth estimation
 - ▣ Detect objects
- ◆ Limitations include poor registration in some scans
- ◆ Future work
 - ▣ Data cleansing/fine-registration before AR measurement
 - ▣ AI method for auto detection



Acknowledgements

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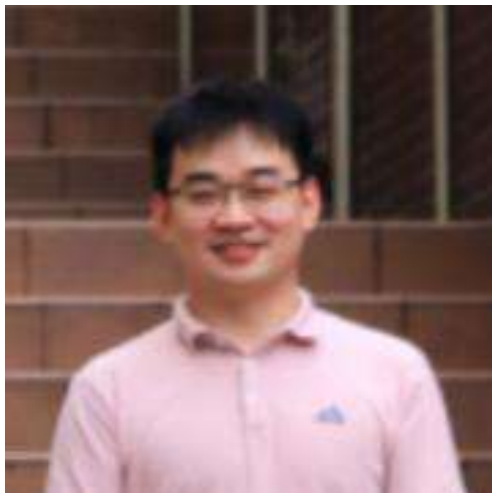
◆ HK RGC General Research Fund (GRF, 17200123)

NDCHK

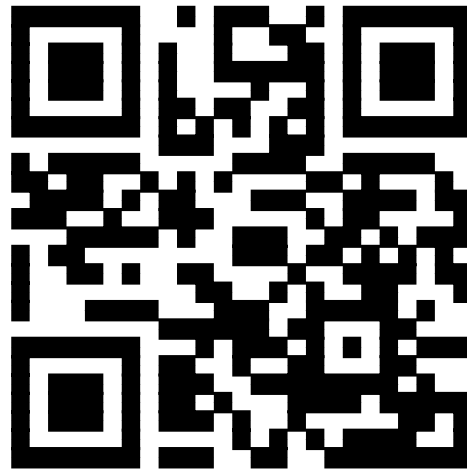
◆ HKU Knowledge Exchange (KE) (KE-IP-2024/25-8),
Teaching Development Grants

◆ Talented students

▣ Meng Sun, Dong Liang, Hank Chen, Longyong Wu



GPR-AR



Scan to explore



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Thank you!

Q&A

